

COLUMN | PRITH BANERJEE & JAMIE J. GOOCH

Daring to Dream of Simulation's Reality in 2041



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It's 2041. Asking your phone to send an autonomous electric vehicle to pick you up for work is common. You still go to work on occasion, but more so for the benefits of face-to-face social interaction than any real need to be in the office. Your home office can access the cloud's essentially unlimited computing power and capacity via 8G, of course, but interacting with holograms of your co-workers and colleagues still isn't quite the same as seeing them in person.

Besides, it's your boss's birthday today. There will be cake.

That reminds you: You wanted to simulate the effect of a significant glucose spike on the artificial pancreas your team is designing. As your AV smoothly navigates the city streets, you call up your simulation software's AI assistant, which greets you by name. You tell it what you

want to do. It asks you a few questions to nail down the requirements and informs you that the new simulation results will be ready before you even get to the office.

"Would you like to 3D print new models using the 10 most promising altered-cell scaffold materials?" it asks.

No need for that yet; the interactive 3D simulations will tell you all you need to know.

Maybe that scenario sounds impossible, but at Ansys we have been singularly focused on helping customers achieve the seemingly impossible for more than 50 years. We've seen what our customers have been able to create with our simulation software, and we know the current incredible rate of innovation today will seem painfully slow two decades from now.

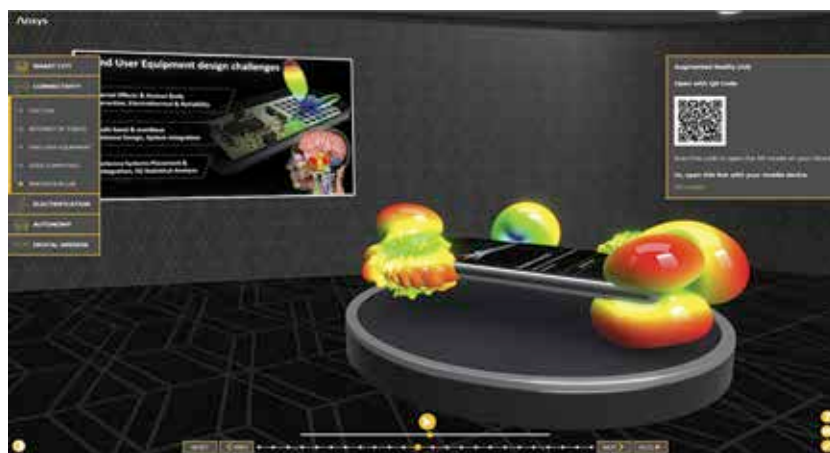
Twenty years ago, when Digit first began publishing, we were all impressed by a mobile phone that had a tiny color screen. We marveled at the first self-contained artificial heart implant. In the simulation world, Ansys 6.0 made large-scale modeling practical for the first

time. Today, we can create 3D scans using lidar sensors in our phones, thousands of artificial hearts are implanted each year and the definition of a "large-scale" model has expanded to include entire systems.

Twenty years from now, as foundational technologies like artificial intelligence/machine learning (AI/ML), high-performance computing (HPC) and augmented reality (AR) simultaneously make huge leaps forward, simulation software will build upon those innovations with complete solutions in different verticals. The ability to simulate what will happen and optimize products, systems and processes before a physical prototype is built will enable us to realize the full promise of today's potentially world-changing innovations: digital twins, personalized medicine, autonomous vehicles, Industry 4.0, smart cities and more.

Simulation Simplified

To understand the future of simulation, you must first understand how simulation software works today. At a very high level: Simulation enables you to predict what will happen to something if it is placed in a given environment under a certain set of circumstances. That "something" could be a soda can being dropped from a shopping cart, a cell phone signal interfering with electronics in a hospital room, or a printed circuit board (PCB) expanding because of the heat of an integrated circuit.



Augmented reality will help us visualize what we cannot see with the naked eye, such as electromagnetic interference.